

ECE 6360 Parallel and Heterogeneous Computing

Homework 3: Cache Planning

Assume that we have a cache with the following parameters:

1. **Total cache size of 64 bytes**
2. **Cache lines of 16 bytes**
(16 bytes are retrieved from DRAM and stored in cache on every miss)
3. **Cache lines are replaced using a LRU policy**
(*least recently used*, the cache line that with the oldest prior access is replaced)

Now assume we want to perform the following matrix multiplication using 32-bit `float` precision:

$$s = Av$$
$$\begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \end{bmatrix} = \begin{bmatrix} a & e & i & m \\ b & f & j & n \\ c & g & k & o \\ d & h & l & p \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

Calculate the cache hit rate (percentage of cache hits for total memory accesses) in two cases:

- The matrix A is stored in column-major format
- The matrix A is stored in row-major format

Keep in mind that both reads and writes will be cached.